Master of Applied Technologies

Course No	COMP8831
Course Name	Machine Learning
Level	8
Credits	15

Student Name(s)	______________________
Student ID(s)	______________________
Assessment Type	Project Proposal Report
Weighting	10%
Due Date and Time	28/08/2026 23:59
Total Marks	100

Student Declaration

I confirm that:

- This is an original assessment and is entirely my own work.
- The work I am submitting for this assessment is free of plagiarism. I have read and understood the Academic Integrity Procedure. I have also read and understood the Student Disciplinary Statue.
- Where I have used ideas, tables, diagrams etc of other writers, I have acknowledged the source in every case.

Students Signature: ______________________

Date: ______________________

1 Assignment Aims

- To propose a machine learning project that integrates the skills and knowledge gained through the course.
- To achieve greater awareness of the latest artificial intelligence tools and techniques for prediction, recognition, optimisation, and decision making.
- To understand the principles, advantages, limitations and possible applications of machine learning algorithms.

2 Assignment Task

What This Proposal Must Deliver

Your proposal is assessed on four things. Make each one easy to find in your report.

1. A clear, real problem that is suitable for a machine learning project.
2. **One modern method per team member**, with **at least one baseline model** to compare against.
3. A specific dataset that you have confirmed is available and appropriate.
4. A realistic plan that can be completed in the time available.

In this assignment, you will write a proposal for a machine learning project that you will later implement as your final project. Think of this proposal as the blueprint for your final project. You are defining the problem, choosing your methods, and planning your approach now so that you have a clear direction when you begin the implementation. A well-written proposal will make your final project significantly easier to execute, so invest the time to plan thoroughly.

You are to search, select, and propose a machine learning project in a written report. Your proposal should be between **1500 and 3000 words** and should not exceed **seven pages**. The word count and page limit both exclude the title page and references. Use font size 11–12, Times family, single spacing.

The proposal weighs **10 percent** of your final mark, so it needs to be written carefully.

You may work **individually or in a group of up to three people**. A group submits **one proposal** that lists the full name and student ID of every member. Every project must include **at least one baseline model** and **one modern (advanced) method per team member**. An individual proposes one baseline and one advanced model, a group of two proposes one baseline and two advanced models, and a group of three proposes one baseline and three advanced models.

Your proposed project should fit supervised, unsupervised, or reinforcement learning domains.

Important Requirements

- Choose a problem that is substantive enough to serve as a capstone ML project. Make sure you can clearly justify its complexity and real-world relevance.
- **Modern methods:** Include **at least one modern, state-of-the-art method for each team member**, for example a neural network, a modern ensemble model, a deep learning architecture, a transformer variant, a graph neural network, or a diffusion model.
- **At least one baseline:** Include **at least one baseline model** and compare it with your modern method or methods. Classical methods such as linear or logistic regression, support vector machines, random forests, and decision trees are welcome as baselines, but must not be your proposed main methods.

3 Scope and Feasibility

Choose a problem that you can realistically complete. Before you submit, confirm that the dataset exists and is accessible, check that its licence allows academic use, and check that training is feasible with the tools and time you have. State any risk to the project and how you plan to manage it.

4 Recommended Models

These are recommendations, not requirements. Choose the methods that fit your problem, and remember that you need **one modern method per member** and **at least one baseline**.

Baseline models
Linear Regression, Logistic Regression, Decision Tree, Random Forest, k-Nearest Neighbours, Support Vector Machine, Naive Bayes.

Modern methods
Convolutional Neural Networks such as ResNet or EfficientNet for images, Vision Transformers, YOLO for object detection, LSTM or GRU networks for sequences, Transformer models for text, XGBoost and LightGBM for tabular data, and diffusion models for generation.

5 Example Project

The example below shows the level of detail expected, here for a group of two. Use it as a guide, not a topic to copy.

Title
Classifying Pneumonia in Chest X-ray Images using Convolutional Neural Networks.

Problem statement (one sentence)
Reading chest X-rays for pneumonia is slow and can vary between clinicians, so an image classifier that flags likely pneumonia could support faster and more consistent screening.

Dataset
Chest X-Ray Images (Pneumonia), Kermany et al. (2018), about 5,863 labelled chest X-rays in two classes, Normal and Pneumonia, available on Kaggle.

Baseline model
A Support Vector Machine on resized, flattened images.

Advanced models (two, one per member)
A transfer-learning CNN, for example ResNet50 fine-tuned on the X-rays, and a Vision Transformer.

Planned comparison
The models will be compared using accuracy, precision, recall, F1 and AUC, and the best model will be recommended.

Expected outcome
A trained pneumonia classifier in which the deep learning models are expected to outperform the baseline, reported with the metrics above and a single recommended model.

6 Proposal Structure

Your proposal must include the following sections in this order.

6.1 Title Page

Include the project title, full name(s) of group member(s), student ID(s), course code, and date.

6.2 Table of Contents

List all sections and subsections with page numbers.

6.3 Abstract *(150–250 words)*

Provide a concise summary of your project. It should cover:

- The problem you intend to solve.
- The machine learning approach you propose.
- The expected outcome.

6.4 Problem Statement

Define the problem your project addresses:

- What is the problem and why does it matter?
- What is the real-world significance or application?
- Why is machine learning a suitable approach for this problem?

6.5 Background / Brief Literature Review

Demonstrate awareness of existing work in your chosen area:

- Review 5–8 key existing works (papers, systems, or methods) related to your problem.
- Summarise relevant approaches that have been applied to similar problems.
- Show how existing work informs your choice of methods and dataset.

6.6 Project Methodology

Describe the technical details of your proposed implementation:

Preferred Learning Methods
Identify the specific ML algorithms or architectures you plan to use. Explain how they work at a high level.

Required Datasets
Specify the datasets you will use for training and testing. Include the source, approximate size, and key features. If using multiple datasets, explain the role of each.

Method Justification
Explain why your chosen methods are appropriate for the problem, and which baseline you will compare against.

Tools and Languages
State your preferred programming language, libraries, and frameworks.

6.7 Expected Outcomes / Deliverables

Describe what your completed project will produce:

- A cleaned and processed dataset ready for modelling.
- More than one trained model applied to the problem.
- Evaluation of each model using appropriate performance metrics.

- A comprehensive comparison of model results.
- A recommended final model with justification for why it is the best fit for the problem.

6.8 References

List all sources cited in your proposal using **IEEE referencing style**. You must include a minimum of **8 quality sources** (peer-reviewed papers, conference proceedings, books, or reputable technical resources). Examples of IEEE formatting can be found at:
`http://libguides.murdoch.edu.au/IEEE/sample`

7 Marking Criteria (100 Marks)

Criteria	Marks	What We Assess
Abstract	5	Clear summary of problem, approach, and expected outcome
Problem Statement	10	Clearly defined problem, real-world relevance, justification for ML
Background / Literature Review	15	Awareness of existing work, relevant sources reviewed
Learning Methods	15	Appropriate modern methods selected, technical understanding demonstrated
Dataset Plan	10	Suitable data identified, source cited, size and features described
Method Justification	15	Why chosen methods fit the problem, rationale for model selection
Expected Outcomes	10	Clear deliverables, baseline and advanced models compared, final model proposed
Writing Quality & Structure	10	Academic tone, logical flow, formatting, grammar
References	10	IEEE style, minimum 8 quality sources, properly cited in text
Total	**100**	

8 Report Format

- Use MS Word or LaTeX for preparing your report.
- Number all sections and subsections.
- Number all pages.
- Ensure the table of contents matches section numbers and page numbers.
- Put captions on all figures, tables, and diagrams.
- Place page breaks appropriately (not immediately after headings or in the middle of a figure/table).
- Cite all work from other sources in the text and list them in your references.
- Most of your report should be written in your own words. Each quotation, table, or diagram from another source must be accompanied by your own discussion of what it shows, why you included it, and why it is relevant.

9 Self-Check Before You Submit

- The four required deliverables above are each easy to find.
- There is **one modern method per member** and **at least one baseline**, and the planned comparison is described.
- The dataset is specific, its source is given, and you have confirmed access.
- The report has all eight sections in order, with numbered sections and pages.
- The length is within **1500 to 3000 words and seven pages**, excluding the title page and references.
- There are **at least eight IEEE references**, each cited in the text.
- Figures and tables have captions.
- The Turnitin similarity, excluding references, is **below 10%**.

10 Submission

Upload your proposal into Moodle through the Turnitin Assignment Submission module (under Assessments, "Project Proposal Submission Link") before the due date. The Turnitin similarity score, excluding references, should be less than **10%**.

Each proposal is reviewed by the lecturer, and individual feedback will be provided. Once the proposal gets approved, the group can start the implementation of the project accordingly.

Policies

Academic Integrity

Marks will be deducted if you fail to draw upon appropriate professional and academic literature or if the assignment is not written and presented in a professional manner. When choosing sources, check that they have passed some form of quality assurance (books, journal articles, and conference articles are a good choice). If it is difficult to identify the author, year, or title of an article then it has not been subject to quality assurance.

Plagiarism will be reported to the Programme Authorities who may award zero marks or refer the matter to the Discipline Committee, which has powers of suspension/exclusion.

Late Submission of Assignments

Assignments submitted after the due date and time without having received an extension through Affected Performance Consideration (APC) will be penalised according to the following:

- **10%** of marks deducted if submitted within 24hrs of the deadline.
- **20%** of marks deducted if submitted after 24hrs and up to 48hrs of the deadline.
- **No grade** will be awarded for an assignment that is submitted later than 48hrs after the deadline.

Assignments submitted in more than 48 hours late will not be marked unless Affected Performance Consideration (APC) apply. So, it is better to submit an incomplete assignment on time.

Affected Performance Consideration

A student, who due to circumstances beyond his or her control, misses a test, final exam or an assignment deadline or considers his or her performance in a test, final exam or an assignment to have been adversely affected, should complete the Affected Performance Consideration (APC) form available from Student Central or online:

`https://www.unitec.ac.nz/current-students/study-support/affected-performance-consideration`

Assistance to Other Students

Students themselves can be an excellent resource to assist the learning of fellow students, but there are issues that arise in assessments that relate to the type and amount of assistance given by students to other students. It is important to recognise what types of assistance are beneficial to another's learning and also what types of assistance are unacceptable in an assessment.

Beneficial Assistance:

- Study Groups.
- Discussion.
- Sharing reading material.
- Testing another student's programming work using the executable code and giving them the results of that testing.

Unacceptable Assistance:

- Working together on one copy of the assessment and submitting it as own work.
- Giving another student your work.
- Copying someone else's work. This includes work done by someone not on the course.
- Changing or correcting another student's work.
- Copying from books, Internet etc. and submitting it as own work.

Do you want to do the best that you can do on this assignment and improve your grades? You could:

- Talk it over with your lecturer
- Visit Student Success and Achievement for learning advice and support
- Visit the Centre for Pacific Development and Support
- Visit the Centre for Maori Development and Support